

Literature Status: Reflexive Infinite-Dimensional Calkin Algebras

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Status

This is a literature-already-answered packet, not an original proof from the run. Problem 1 in Motakis–Puglisi–Tolias, arXiv:1711.01340, asks whether there exists a Banach space whose Calkin algebra is reflexive and infinite dimensional. Motakis–Pelczar-Barwacz, arXiv:2401.18037, answers this affirmatively.

Original Problem

The original source is:

P. Motakis, D. Puglisi, A. Tolias, *Algebras of diagonal operators of the form scalar-plus-compact are Calkin algebras*, arXiv:1711.01340.

In the final remarks section, Problem 1 asks:

Does there exist a Banach space the Calkin algebra of which is reflexive and infinite dimensional?

X^* does not embed in an L_1 space then $\mathcal{V}_X^*$ is not a $\mathcal{L}_1$ -space. Such a space is, e.g., $X = J^* = \mathcal{J}_*(\ell_2)$ the dual of which is isomorphic to J which has trivial cotype. In fact, with some work, something stronger can be shown: if $\mathcal{V}_X$ is a $\mathcal{L}_\infty$ -space then so is X .

Remark 8.3. The space $\mathcal{V}_X$ has the bounded approximation property. To see this recall that $(Z_n)_n$ is a Schauder decomposition of $\mathcal{V}_X$ and the spaces $\sum_{k=1}^n \oplus Z_k$ are uniformly isomorphic to $((\sum_{i=1}^n \oplus X_k)_{\text{utc}} \oplus \ell_\infty(\Gamma_n))_\infty = (\mathcal{X}_n \oplus \ell_\infty(\Gamma_n))_\infty$. Recall that each X_k has a 2-Schauder basis $(t_{k,i})_i$. Using the definition of $\mathcal{G}^{\text{utc}}$ one can see that the finite dimensional subspaces F_m spanned by $((t_{k,i})_{i=1}^m)_{k=1}^n$ are increasing, uniformly complemented in $\mathcal{X}_n$, and their union is dense $\mathcal{X}_n$.

Remark 8.4. When a space X has a finite dimensional Calkin algebra or one that is isomorphic to ℓ_1 then $\mathcal{K}(X)$ is complemented in $\mathcal{L}(X)$. In the examples from [MPZ], that have infinite dimensional $C(K)$ spaces as Calkin algebras, the space $\mathcal{K}(X)$ is not complemented in $\mathcal{L}(X)$. Regarding the space $\mathcal{V}_X$ this is not always clear. If $\mathcal{K}_{\text{diag}}(X)$ is isomorphic to ℓ_1 then $\mathcal{K}(\mathcal{V}_X)$ is complemented (this happens, e.g., when X is c_0 endowed with the summing basis). On the other hand, if $\mathcal{K}_{\text{diag}}(X)$ contains c_0 then the argument used in [MPZ] goes through and $\mathcal{K}(\mathcal{V}_X)$ is not complemented in $\mathcal{L}(\mathcal{V}_X)$.

Remark 8.5. The construction of spaces with quasi-reflexive Calkin algebras is a step towards trying to find a space with a reflexive and infinite dimensional Calkin algebra. One way for this to be possible would be to find a space X with $\mathcal{L}(X)$ reflexive. As it was pointed out to us by J. A. Chávez-Domínguez, [B, Corollary 2] implies that such an X must be finite dimensional so this route would be a dead end.

Problem 1. Does there exist a Banach space the Calkin algebra of which is reflexive and infinite dimensional?

Remark 8.6. Given a Banach space X with a basis we have used the Argyros-Haydon scheme for defining spaces with the scalar-plus-compact property to obtain a Calkin algebra that is isomorphic as a Banach algebra to the space $\mathbb{R}I \oplus \mathcal{K}_{\text{diag}}(X)$. It is conceivable that one may use the Gowers-Maurey scheme from [G] for constructing a space with an unconditional basis with the “diagonal plus strictly singular” property to construct a space with the property that algebra $\mathcal{L}(X)/\mathcal{SS}(X)$ is isomorphic as a Banach algebra to the whole space $\mathcal{L}_{\text{diag}}(X)$. If one would like to have a space with Calkin algebra $\mathcal{L}_{\text{diag}}(X)$, then a new scheme would be necessary, one that is used to define Banach space with an unconditional basis with the diagonal-plus-compact property.

Problem 2. Does there exist a Banach space with an unconditional basis so that every bounded linear operator on that space is the sum of a diagonal operator with a compact operator?

Problem 3. Let X be a Banach space with a Schauder basis $(e_i)_i$. Does there exist a Banach space Y the Calkin algebra of which is isomorphic, as a Banach algebra, to $\mathcal{L}_{\text{diag}}(X)$?

Remark 8.7. Recall that for all spaces X with an unconditional basis $\mathcal{L}_{\text{diag}}(X)$ is isomorphic as a Banach algebra to ℓ_∞ with pointwise multiplication. As it was explained earlier, a positive answer to Problem 2 could perhaps yield a positive answer to Problem 3 and hence also a positive solution to the following.

Later Answer

The supporting source is:

P. Motakis, A. Pelczar-Barwacz, *Reflexive Calkin algebras*, arXiv:2401.18037.

The abstract states that $\text{Cal}(X)$ can be an infinite-dimensional reflexive Banach space, even isomorphic to a Hilbert space. The main theorem proves more generally that if U has a normalized 1-unconditional basis and no c_0 asymptotic version, then there is a Banach space $\mathfrak{X}_U$ such that

$$\text{Cal}(\mathfrak{X}_U)$$

is isomorphic, as a Banach algebra, to the unitization of U with coordinate-wise multiplication.

REFLEXIVE CALKIN ALGEBRAS

PAVLOS MOTAKIS AND ANNA PELCZAR-BARWACZ

ABSTRACT. For a Banach space X denote by $\mathcal{L}(X)$ the algebra of bounded linear operators on X , by $\mathcal{K}(X)$ the compact operator ideal on X , and by $\mathcal{Cal}(X) = \mathcal{L}(X)/\mathcal{K}(X)$ the Calkin algebra of X . We prove that $\mathcal{Cal}(X)$ can be an infinite-dimensional reflexive Banach space, even isomorphic to a Hilbert space. More precisely, for every Banach space U with a normalized unconditional basis not having a c_0 asymptotic version we construct a Banach space $\mathfrak{X}_U$ and a sequence of mutually annihilating projections $(I_s)_{s=1}^\infty$ on $\mathfrak{X}_U$, i.e., $I_s I_t = 0$, for $s \neq t$, such that $\mathcal{L}(\mathfrak{X}_U) = \mathcal{K}(\mathfrak{X}_U) \oplus [(I_s)_{s=1}^\infty] \oplus \mathbb{C}I$ and $(I_s)_{s=1}^\infty$ is equivalent to $(u_s)_{s=1}^\infty$. In particular, $\mathcal{Cal}(\mathfrak{X}_U)$ is isomorphic, as a Banach algebra, to the unitization of U with coordinate-wise multiplication. Banach spaces U meeting these criteria include ℓ_p and $(\oplus_n \ell_\infty^n)_p$, $1 \leq p < \infty$, with their unit vector bases, L_p , $1 < p < \infty$, with the Haar system, the asymptotic- ℓ_1 Tsirelson space and Schlumprecht space with their usual bases, and many others.

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Taking $U = \ell_2$ gives an infinite-dimensional reflexive Banach algebra. Therefore the original 1711.01340 problem has an affirmative answer in the later literature.

Limitations

This packet only records the status of the reflexive infinite-dimensional Calkin algebra problem. It does not address the other problems in the same section of arXiv:1711.01340, including the diagonal-plus-compact problem for spaces with unconditional bases or the uncountable $C(K)$ Calkin-algebra problem.

Review Recommendation

Verify that the supporting theorem is being used only through its stated Banach-algebra isomorphism and that the choice $U = \ell_2$ satisfies the hypotheses of the theorem. This is enough for the original question because it only asks for reflexivity and infinite dimensionality of the Calkin algebra as a Banach space.